Student Number:									

3 Complex Analysis 2023

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Test 1

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Instructions:

- This question paper consists of **5 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	Σ	Mark
Maximum Points	12	10	10	10	40	100%
Result						

Student Number:									

Problem 1

(12 points)

(a) Let $G \subseteq \mathbb{C}$ be an open set and $f : G \to \mathbb{C}$. Let $z \in \mathbb{C}$. Define what is meant by f being differentiable at z, and define the derivative of f at z.

(b) Show by definition that the function $f(z) = |z|^2$ is differentiable at $z = 0$, and determine $f'(0)$. (Hint: the definition of the limit can be helpful in considering the limit here.)

(c) Show by a method of your choice that $f(z) = |z|^2$ is not differentiable at any $z \neq 0$.

Student Number:									

Problem 2

(10 points)

(a) Determine the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n} x^{2n}$.

(b) Find the power series expansion of the form $\sum_{n=0}^{\infty} a_n z^n$ for the function $f(z) = \frac{z}{2+z^2}$, and determine its domain of convergence. (Hint: you may use without proof known facts about known series.)

Student Number:									

Problem 3

(10 points)

Recall that $\cos z = \frac{e^{iz}+e^{-iz}}{2}$, $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$, $z \in \mathbb{C}$.

(a) Prove that $\cos^2 z + \sin^2 z = 1$.

(b) Show that the function $\sin z$ is unbounded on $\mathbb{C}$.

(c) Recall that $\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$. Show that this series converges absolutely and uniformly on any closed disc $\{z \in \mathbb{C} : |z| \leq r\}$, where $r > 0$.

Student Number:									

Problem 4

(10 points)

Consider the path

$$\gamma(t) = \begin{cases} t, & t \in [0,1], \\ e^{i\pi(t-1)}, & t \in [1,2]. \end{cases}$$

(a) Sketch the path in the graph paper below. Don't forget to label the axes and to show the orientation of the path.

(b) Calculate $\int_\gamma z^2 \, dz$.